SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1001	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	25/10/2026	**Duration:**	60 minutes

Code of Conduct

I Divya Dharshini V S (192511308) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. Study and Analyze the Scenario

1.1 Carefully understand the given problem statement.

Within this project I hope to create an AI-Based Network Intrusion Detection and Threat Monitoring System that can detect suspicious activities within the network to aid Administrators when trying to take action. The issues this system can help with are:

- Unauthorized Access – Someone wants to get onto the network to steal valuable data
- Suspicious Activities – Someone trying to find open ports, guess passwords and send malicious files

- Time has to be spent looking into the activities
- Detection – Cannot detect new attacks, only previous patterns of attacks.
- Delayed Detection – If attacks aren't detected then there can be data loss.

1.2 Identify the business domain, stakeholders, existing challenges, and system objectives.

Key Stakeholders:

- Security Administrator – Will look at the alerts and find suspicious activities.
- Network Administrator – Manages the network and resolves any connectivity issues.
- Cyber Security Team – Will look at any potential attacks.
- System Administrator – Makes sure the application is working well and manages access.
- Employees/Users – Will use the network to complete tasks and want to keep their data safe.
- Management – Will look at the reports and see if they need more security.

Issues:

- High Traffic – Too much traffic may be occurring so difficult to look at all activities.
- False positives – Some activities may be found as attacks.
- New methods – Attackers may find new ways to avoid detection.
- Too many alerts – Too many alerts could be given which can be confusing.
- Data Security – Data within the network needs to be secure.

Objectives:

- Monitor traffic
- Use AI and Machine Learning to detect suspicious activities.
- Demonstrate how serious potential threats are.
- Clear results.
- Alerts and recording data.
- Help Administrators prioritise threats.

Overall during this project I will aim to make it easier to detect any potential attacks. Although the system may not always detect attacks correctly so needs to be regularly monitored. By improving the model and using good data this should be a useful tool for network security.

2. Identify Functional and Non-Functional Requirements

2.1 List all functional requirements required by the proposed system.

These specify the functionality provided by the system. My system will monitor all data that occurs on a computer network, which would cause an alert to be generated if anything unusual was happening. This will help the admin understand what is going on within the system.

- Monitor Data: This will monitor all data within the network.
- Collect Data: This will collect data.
- Detect Attack: This uses AI to detect any possible attacks.
- Attack Info: This will show information on any possible attacks.
- Alert: This will show an alert.
- Dashboard: This will show information on the network, alerts and results.
- Save Data: This will save previous alerts and results.
- Report: This will generate reports on the network and possible attacks.
- Login: This allows authorised users to login.
- Save Data: This saves important data.

2.2 Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

This will show how well the system will work. I want the system to be easy to use, fast and secure. Below are the key requirements.

- Performance: Results will be shown after analysing data quickly.
- Security: Data will be kept safe and only accessible to authorised users.
- Reliable: The system will work properly and provide useful results.
- Usable: Dashboard will be easy to use with easy to understand results.
- Scalable: The system will be scalable to extract more data from the network.
- Maintainable: The system will be easy to update and any issues will be resolved.
- Available: The system will be available.

3. Identify Actors and Use Cases

3.1 Identify all primary and secondary actors interacting with the system.

Actors represent the users or systems that will interact with the project. The project is an AI Based Network Intrusion Detection and Threat Monitoring System which helps identify potential attacks on a computer network. There are two different types of actors within this project. A primary actor will directly use the system and a secondary actor will help ensure the system works correctly.

The main actors within the project are:

Primary Actors:

- Security Administrator – Will use the system to view alerts and identify potential attacks within the network.
- Network Administrator – Will view network activities and investigate any issues within the network.
- System Administrator – Will manage the application and user access.

Secondary Actors:

- AI Detection Model – Will analyse network data and identify potential attacks.
- Database – Will store network data, alerts and historical data.
- Network Monitoring Tool – This will provide data to the system.

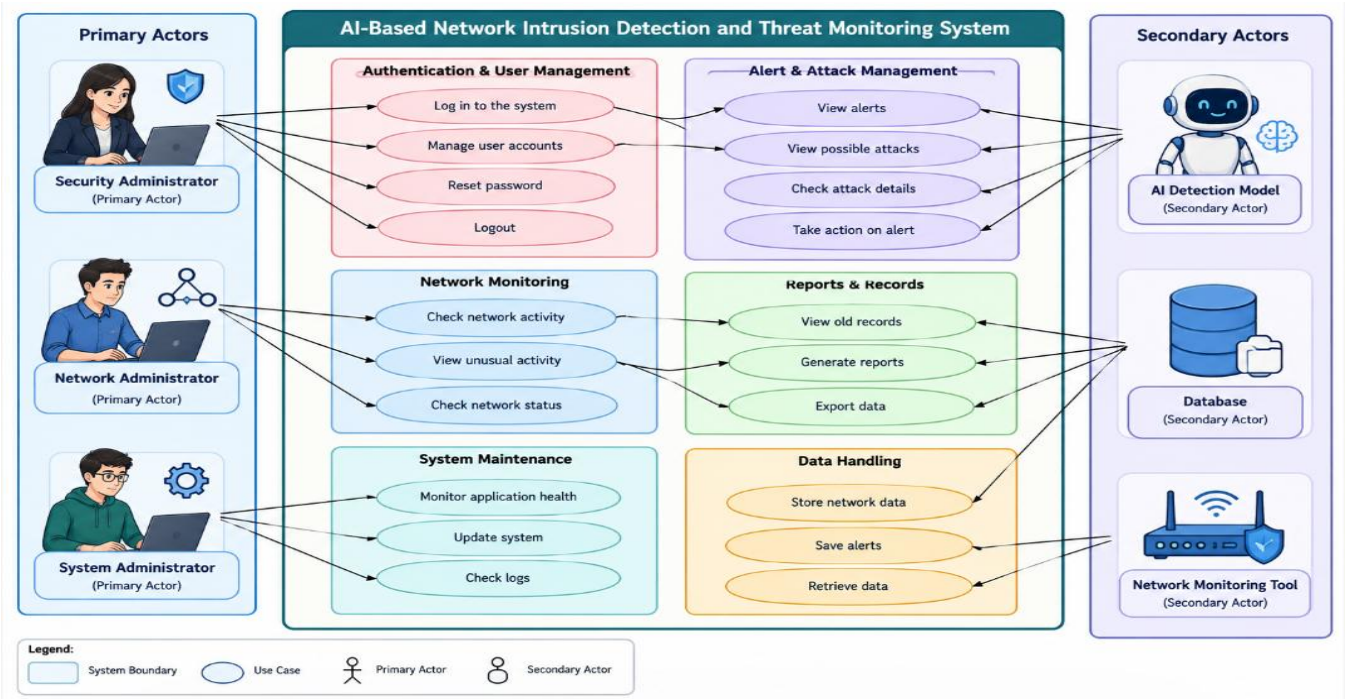
3.2 List the major use cases associated with each actor.

The main use cases for each actor are:

- Security Administrator – Log in, View Alerts, View Potential Attacks, View Historical Data.
- Network Administrator – View network activities, View anomalous activities, View attack details.
- System Administrator – Manage user access, View if the application is working correctly.
- AI Detection Model – Analyse network data, Identify potential attacks.
- Database – Store network data, Store Alerts, Store historical data.
- Network Monitoring Tool – Gather details about network activities and send to system.

An example would be if the system identifies an anomaly, it will display an alert on the screen for the security admin. They will then be able to look at the details and decide what they want to do. They can also look at historical data to see what happened previously. This will help to give an understanding of how they will interact with each other and what features I will need to include within my project.

3.3 Draw a Use Case Diagram representing the interactions between actors and the system.



4. Prepare the Software Requirement Specification (SRS)

Develop an SRS document containing:

4.1 Introduction

To build an AI Based Network Intrusion Detection and Threat Monitoring System to detect potential attack.

4.2 Problem Description

Computers networks are used everyday to share information. Sometimes this information can be accessed by people that shouldn't be able too. All information has to be manually looked through to detect potential attacks, this takes a long time. I will develop a system that detects potential attacks.

4.3 Scope of the System

The system will look through data in the network to detect potential attacks. If unusual information is found an alert will be made. Results will be displayed on a dashboard and past results stored.

4.4 Functional Requirements

- Look through data in the network.
- Use AI to detect potential attacks.
- Alerts will be shown if unusual information is found.
- Results will be shown on a dashboard.
- Past results will be stored and reports generated.
- Login available.

4.5 Non-Functional Requirements

- Fast results.
- Data is secure.
- Reliable.
- User friendly.
- Scalable.

4.6 Use Case Descriptions

- Security Administrator: Can see alerts and potential attacks.
- Network Administrator: Can see network and detect issues.
- System Administrator: Can manage users and see application.

Alert will be shown if unusual information is found. Security administrator will then see this and make a decision.

4.7 Data Requirements

Data from network will be required to detect potential attacks. Alerts and past results will be stored so administrator can see. Data needs to be secure.

4.8 External Interface Requirements

The system will have an easy to use dashboard. Information will be clearly shown relating to network, alerts and results. Data may come from a network monitoring tool.

4.9 System Features

The system will analyse network activity, detect potential attacks using AI and display alerts. Dashboard will be included to display results. Past results will be stored and reports generated to assist the administrator.

5. Identify Assumptions and Constraints

5.1 List the assumptions considered while developing the system.

a) Data can be collected from network traffic.

b) Data will be available with enough information for the AI model to pick up on both normal and suspicious activity.

c) Security and Network Administrators will have some knowledge of computers and dashboards.

d) An internet or network connection will be available.

e) The AI model will be able to detect suspicious activity in the network from the training data but may not be able to detect all attacks correctly.

f) All users will have permission to access the system.

g) Python, Streamlit, Pandas, Scikit-learn and SQLite will be available.

h) Regular updates to the system will be made if new types of attacks are detected, security issues arise or attacks can be picked up more efficiently.

5.2 Identify technical, operational, economic, legal, and time-related constraints affecting the project.

a) Technical Constraints

This system will need a computer, software, network data and an AI model. If there is a lot of network traffic this could cause delay. The training data could be inaccurate or incomplete causing accuracy issues. This system may also need extra configuration and permission to connect to real network devices.

b) Operational Constraints

This system will need to be properly managed by authorised users. Administrators will need to know what alerts mean and check any suspicious activity. There could be a lot of false alerts causing confusion. This system may need to be monitored regularly, updated and records checked.

c) Cost Constraints

This project will need to be developed within a budget. Using paid security software, high-end hardware, cloud and professional software may cause the cost to go up. This project will mostly use free tools such as Python, Streamlit, Pandas, Scikit-learn and SQLite to keep the cost down.

d) Legal Constraints

This system will need to consider privacy and data protection. The system should not be looking at private information without permission. The network information being monitored should be done according to the organisations policies and any relevant laws. Only authorised users should be able to see sensitive records.

e) Time Constraints

This project will need to be completed within the time given by the college. This project will take time to plan, code, test, create the AI model, design the dashboard and document. There may not be much time so fewer features can be added.

6. Validate Requirement Completeness and Consistency

6.1 Verify whether all stakeholder requirements have been addressed.

Different stakeholders will have different requirements.

- Security Administrators want the ability to detect potential attacks, receive alerts and view historical security data. Our system will provide this.
- Network Adminstrators want to analyse network traffic and identify anomalies. This can be achieved by using the network traffic analysis feature.
- System Administrators want to manage user access to the application.
- Management want to know what issues they are having with their network security. They can do this by viewing reports and storing data.

6.2 Check for ambiguity, redundancy, inconsistency, and missing requirements.

a) Ambiguity

A requirement that is ambiguous does not make sense. An example of this would be if the requirement says ‘The system should quickly detect attacks’. It doesn’t say how quickly it should detect attacks.

Improvement: The requirements should specify a suitable amount of time for detecting attacks and displaying alerts.

b) Overlapping

Overlapping occurs when you mention the same thing more than once. For example identifying suspicious activities and identifying unusual activities may be very similar.

Improvement: Ensure you merge similar requirements to make one clear requirement.

c) Conflicts

Conflicts occur when there are 2 requirements that conflict with each other. For example one requirement could state that everyone can view security data whilst another could state that only authorised users can view this.

Improvement: The system should have clear guidelines on who can see the data. Only users with permission should be able to view key security data.

d) Missing requirements

A missing requirement is when the feature isn't mentioned. For example detecting an attack may be mentioned but there is no mention of how the alert will be stored/analysed.

Improvement: Key features should be included such as storing alerts, analysing historical data, user management, error checking and analysing false alerts.

6.3 Suggest improvements to enhance the quality and completeness of the requirement specification.

Below are some recommendations to improve the requirements of my project.

- Use simple language – All requirements should clearly state what the system should do.
- Be precise – Any requirements regarding speed and accuracy should have suitable values to test.
- All key features – Clearly state that the system has network monitoring, attack detection, alerts, dashboards, reports, logins and data storage.
- Improved security – Ensure that the systems keeps the networks data secure and only allows authorised users.
- Test requirements – Ensure all requirements are tested to ensure the feature works as intended.
- Feedback – Get feedback from administrators to make any required changes.
- Update – If there are any new security requirements then update the requirements.
- Link to testing – Ensure each requirement links to the correct feature and test case.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25